

Answer key (facilitator copy)

Question 1. All three equations produce identical predictions: 9, 7, 17, 11, 17. Every error is 0, so $\text{SSE} = 0$ for A, B, and C.

Question 2. All of them. Each is a perfect fit, so there is no single best equation.

Question 3. The columns are linearly dependent: $x_3 = x_1 + x_2$ in every row ($3 = 1 + 2$, $3 = 2 + 1$, $6 = 3 + 3$, $5 = 4 + 1$, $7 = 5 + 2$). So x_3 carries no information that x_1 and x_2 do not already carry, and any weight placed on x_3 can be traded for weight on x_1 and x_2 . For example, going from A to B moves 1 unit of weight onto x_3 and takes 1 unit off each of x_1 and x_2 , which changes nothing because x_3 is exactly $x_1 + x_2$.

Question 4. The coefficient on x_4 is 1 in all three equations, while the coefficients on x_1 , x_2 , x_3 all change. The column x_4 is not a combination of the others, so nothing can be traded for it: any change to its weight makes the predictions worse. Only the dependent columns have freedom.

Question 5. Any equation of the form

$$\hat{y} = (2 + t)x_1 + (3 + t)x_2 - tx_3 + 1x_4$$

works. A is $t = 0$, B is $t = -1$, C is $t = -2$. A new one: $t = 1$ gives $\hat{y} = 3x_1 + 4x_2 - 1x_3 + 1x_4$. Check row 0: $3(1) + 4(2) - 3 + 1 = 9$. Check row 3: $3(4) + 4(1) - 5 + 0 = 11$. Both match. Students who change the x_4 coefficient should find the check fails.

Question 6. Infinitely many, one for every real value of t . The question “what are the coefficients?” has no unique answer when columns are linearly dependent. The data cannot distinguish between the solutions, so the choice among them comes from outside the data (a regularizer, a pseudoinverse convention, or dropping a column).

Discussion points.

- The design matrix X has rank 3 but 4 columns, so $X^T X$ is singular and the normal equations have a whole line of solutions rather than one. The line runs in the direction $(1, 1, -1, 0)$, which is why b_4 is pinned down while b_1, b_2, b_3 are not.
- The smallest singular value of X is exactly 0 in theory. In floating point it shows up as roughly 10^{-16} , so `np.linalg.cond(X)` reports something like 10^{16} instead of infinity. This is the same “high condition number” warning that statsmodels prints.
- Dropping x_3 fixes it: the remaining columns x_1, x_2, x_4 have rank 3 and a condition number of about 16, and the unique best fit is $\hat{y} = 2x_1 + 3x_2 + 1x_4$.
- Real data is rarely this exact, but near-dependence behaves similarly: many coefficient vectors fit almost equally well, and the coefficients become unstable.